

SONIKA SRINIVAS

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SUMMARY

Information Systems graduate with 2+ years of experience in Software & data engineering, quantitative analytics, and machine learning, building reproducible data pipelines, analytical platforms, and AI-enabled decision-support solutions for large-scale datasets. Experienced in outcomes measurement, indicator analysis, data quality, evidence-based evaluation, and results reporting. Strong technical foundation in Python, SQL, machine learning, LLMs, cloud data platforms, statistical analysis, and data visualization, with experience translating complex analytical findings into reliable dashboards and actionable insights. Combines software engineering rigor with analytical and research capabilities to connect data, evidence, and operational decision-making across complex, multidisciplinary environments.

EDUCATION

San Diego State University, San Diego, CA

Aug 2025 - May 2027

Master of Science in Information Systems | GPA: 3.5/4.0

Coursework: Advanced Python Programming, Machine Learning, Statistical Analysis, Database Management, AI & Big Data, Time series Analysis, Project Planning & Management.

Visvesvaraya Technological University, Bengaluru, India

Aug 2019 - Aug 2023

Bachelor of Engineering in Information Science | GPA: 3.5/4.0

Coursework: Data Structures and Algorithms, Cloud Computing, Big Data Analytics, AI and ML, Mathematics (Stats & Calculus).

TECHNICAL SKILLS

Programming: Python, SQL, Java, JavaScript, R, sPySpark, Windows PowerShell.

Data Engineering & Big Data: Apache Spark, Kafka, Databricks, Docker, Snowflake, Hive

Databases: SQL Server, CosmosDB NoSQL, PostgreSQL, MySQL, MongoDB, Redis.

Agentic AI & LLMs: LangChain, GitHub Copilot, Claude, OpenAI API, Groq LLM.

Cloud Platforms: Microsoft Azure (Data Lake, Data Factory, Synapse, Functions, DevOps, Cosmos DB), AWS, Google Cloud Platform (GCP), OCI (Oracle Cloud).

Machine Learning: Scikit-Learn, TF-IDF, SVM, Random Forest, XGBoost, PyTorch, LightGBM.

Full-Stack Development: Vue.js, Node.js, Nginx, REST API, Flask API, HTML, CSS.

DevOps & Version Control: Git, GitHub, GitHub Actions, CI/CD.

Data Visualization & Tools: Tableau, Power BI, Streamlit, Google Analytics, Excel, MS Suite.

Certifications: Microsoft Azure Fundamentals (AI-900).

EXPERIENCE

Data Analyst Student Assistant | San Diego State University, San Diego, USA

Feb 2026 - Present

- Architected and deployed an OpenAI API-driven applicant screening workflow with built-in auditability guardrails, reducing manual review time by 83% while maintaining strict data governance standards in a compliance-sensitive academic environment.
- Engineered prompt design and validation logic to ensure reproducibility, traceability, and explainability of AI-generated screening decisions, supporting auditability requirements in a governance-heavy research setting.
- Built automated data quality and consistency checks for incoming applicant datasets, ensuring reliable, standardized inputs to the AI screening pipeline and minimizing downstream errors.

Software Development Engineer | Ericsson, Bengaluru, India

Sept 2023 - Jul 2025

- Architected end-to-end migration from on-premise MySQL to Azure Cosmos DB, designing partition key strategy, NoSQL schema, and indexing to eliminate cross-partition queries cutting latency by 74% and cost per request by 30%; built Azure Data Factory pipelines and Azure Functions for ETL orchestration across millions of records.
- Refactored 2,500+ SQL ETL queries into optimized Cosmos DB NoSQL access patterns; re-engineered data ingestion and transformation logic with schema validation and error handling at each stage, increasing system throughput 4x.
- Built and owned CI/CD pipelines (GitHub Actions, Azure DevOps) for automated deployment of data pipelines, implementing staged rollouts and automated integration tests achieving zero-downtime releases and reducing deployment incidents by ~90%.

Software Development Engineer Intern | Ericsson, Bengaluru, India

Feb 2023 - Sept 2023

- Developed and delivered end-to-end features for an internal platform as a full-stack developer, building responsive Vue.js frontends tightly integrated with backend services and REST APIs — boosting team efficiency by 30% through streamlined, reusable workflows. Optimized backend data processing pipelines and API-driven data flows, cutting data processing time by 40% through query tuning, caching strategies, and more efficient service-layer logic.

- Improved system reliability and accelerated feature delivery by architecting modular, decoupled components across the stack and enforcing clean, well-documented REST API contracts between frontend and backend teams.

PROJECTS

Personalized Content Recommender | *PyTorch, LightGBM, FastAPI, scikit-learn, scipy, Pandas*

- Task: Popularity-based ranking served identical content to all users regardless of click history. Designed and implemented a two-stage ML pipeline on the MIND dataset (2.7M impressions, 50K users, 22K articles) to replace it with personalized recommendations.
- Action: Built a PyTorch Two-Tower network (dual-encoder MLP, 64-dim embeddings, BCEWithLogitsLoss, 945K balanced training samples) for top-100 candidate retrieval, followed by a LightGBM re-ranker trained on 5 behavioral features.
- Result: A/B tested across 47,000 users LightGBM re-ranker achieved CTR@5 of 0.1428, a 38.8% lift over the popularity baseline ($p < 0.000001$), served via a stateless FastAPI REST endpoint with sub-100ms p99 latency.

Real-Time Clickstream Analytics Platform | *Apache Kafka, Faust, PostgreSQL, Redis, Grafana, Docker*

- Task: E-commerce platforms lack real-time visibility into funnel drop-off, active sessions, and traffic spikes. Architected a production-grade streaming pipeline to process raw user interaction events end-to-end in under 2 seconds.
- Action: Built a Faust stream processor on a 3-partition Kafka topic (keyed by user_id for session locality) implementing tumbling, sliding, and funnel-step aggregation pipelines with 30-second watermark tolerance for late/out-of-order events.
- Result: Dual-sink architecture writes to PostgreSQL (idempotent upserts) and Redis sorted sets (sub-1ms reads); processes 500 events/second at under 2 seconds end-to-end latency, fully containerized via Docker Compose.

Insurance Premium Predictor | *Python, Random Forest, SHAP*

- Task: Engineered an end-to-end ML pricing pipeline on 5,800+ insurance records to predict premiums while addressing model accuracy, feature interactions, interpretability, and fairness.
- Action: Designed domain-driven interaction features (bmi_smoker, age_smoker, age_bmi) and trained a Random Forest regression model; integrated SHAP TreeExplainer for per-prediction attribution and cohort-level fairness analysis across sex and smoker groups; deployed the production-ready pipeline to the cloud.
- Result: Increased baseline Linear Regression R^2 from 0.750 to 0.866 through feature engineering and achieved $R^2=0.875$ with RMSE=4,406 using Random Forest, while providing interpretable predictions and identifying no systematic demographic bias across evaluated cohorts.

PUBLICATIONS

The Personalization Paradox: A Bias-Enabled Guidance Trap

Working paper, SSRN, December 2025

Examined how AI-driven career coaching systems, in optimizing for personalization through relevance weighting and reinforcement loops, can overfit to biased demographic patterns and narrow the aspirational paths shown to users. Conducted interviews with AI system architects to identify design mechanisms including diversity constraints, exploration parameters, and transparency features that balance algorithmic optimization against user autonomy. Proposed the "Epistemic Immune System" (EIS), a framework treating bias as a diagnostic signal rather than noise, using controlled bias exposure, scenario simulation, and structured exploration to surface hidden bias in AI-guided decision-making.

Available at SSRN: [The Personalization Paradox: A Bias-Enabled Guidance Trap](#)